

Machine Learning Techniques for Cross-Sectional Equity Returns' Prediction

Data:

Feature	Type	Expected sign	How the paper built it	US source fields
Beta	Market risk	(paper found negative)	CAPM beta from daily returns over prior 12 months	Price series (SEP) + a market index
Size	Market cap	(paper found ~none)	Log of market equity	shares outstanding × price (SEP)
Book-to-market (be2me)	Value	positive	Book equity ÷ market equity	book equity (SF1) ÷ market cap
Momentum (mom)	Trend	positive	Total return over prior 12 months, skipping the most recent month (i.e. months t-12 to t-2)	Price series (SEP)
Investment (inv)	Asset growth	negative	Year-over-year growth in total assets	total assets (SF1)
Operating profitability (op)	Quality	positive	Operating income ÷ book equity	operating income (SF1) ÷ book equity (SF1)

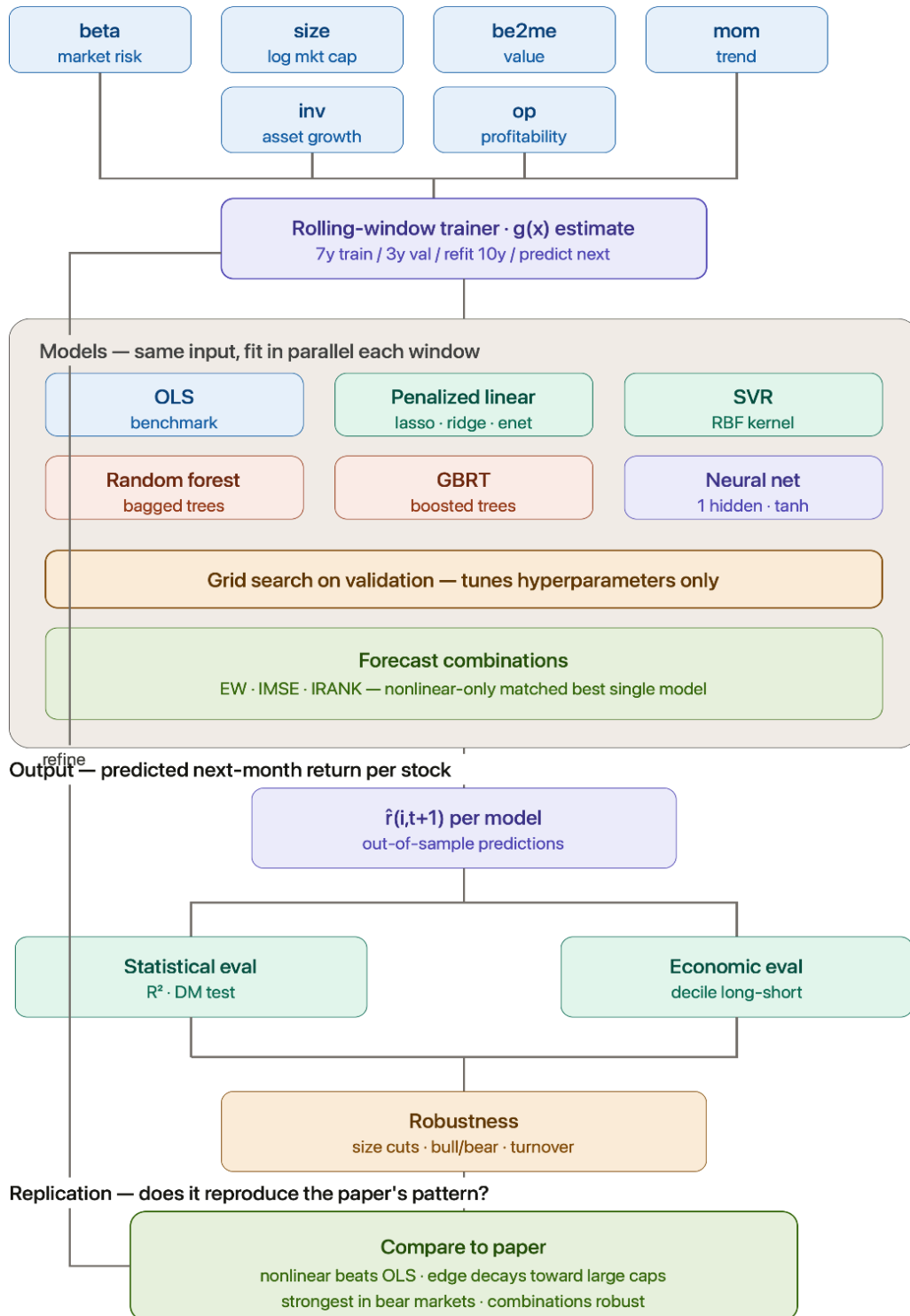
Lag the accounting data. The paper assumes accounting numbers become public **six months after fiscal year-end** to avoid look-ahead bias. So when predicting returns in month t , you only use fundamentals from a fiscal period that ended at least six months earlier. This is the single most important rule for a tradable build — get it wrong and your backtest is fiction.

Lag the price-derived features too. Beta uses the *previous* 12 months. Momentum *excludes the last month* (the skip-month is deliberate — it avoids short-term reversal contaminating the momentum signal). Everything is known strictly before the prediction month.

Rank-normalize each feature cross-sectionally each month, mapped into roughly $[-1, 1]$. This handles different scales (market cap in millions vs. a ratio near 1) and crushes outliers. Do it per-month, across the cross-section, not pooled across time.

Missing-value handling. The paper's whole pitch is that they *chose* six predictors available across the full period so they rarely have to impute. When a characteristic is missing but the return exists, they fill with the **cross-sectional median** (which after rank-normalizing is ~ 0). They explicitly *avoid* filling entire cross-sections with zeros, which is the bad practice they criticize in prior work. Copy this: only impute sparingly, never backfill whole columns.

Input — six firm characteristics $x(i,t)$, rank-normalized per month



- Same six-feature input flows to every model — the only thing that varies is $g(x)$
- Replication confirmed when your pattern matches the paper's; then deviate for tradability
- Left loop: if pattern doesn't match, the bug is usually in features or training — refine and rerun

The data they used here is the European stock data. If we need to use it for ourselves we need to use the USA stock data and we can't only have the S&P500. The ML edge will evaporate with the large 500 companies.

Sharadar (via Nasdaq Data Link). — For data.